

Section 12.3: The Dot Product

Vectors and the Geometry of Space

MTH 310
Calculus III

The Dot Product

We can **add** vectors and **scale** them. But can we "multiply" two vectors?

The **dot product** takes two vectors and gives a **scalar** (a number).

Definition 1: Dot Product

If $\vec{a} = \langle a_1, a_2, a_3 \rangle$ and $\vec{b} = \langle b_1, b_2, b_3 \rangle$, then

$$\vec{a} \cdot \vec{b} = a_1 b_1 + a_2 b_2 + a_3 b_3$$

Multiply corresponding components and add.

Computing Dot Products

Example 1

Compute each dot product.

$$\langle 2, 4 \rangle \cdot \langle 3, -1 \rangle$$

$$\langle 2, 4 \rangle \cdot \langle 3, -1 \rangle = 2(3) + 4(-1) = 6 - 4 = 2$$

$$\langle -1, 7, 4 \rangle \cdot \langle 6, 2, -\frac{1}{2} \rangle$$

$$\langle -1, 7, 4 \rangle \cdot \langle 6, 2, -\frac{1}{2} \rangle = -6 + 14 - 2 = 6$$

$$(\vec{i} + 2\vec{j} - 3\vec{k}) \cdot (2\vec{j} - \vec{k})$$

$$\langle 1, 2, -3 \rangle \cdot \langle 0, 2, -1 \rangle = 0 + 4 + 3 = 7$$

$$\vec{a} \cdot \vec{a} = a_1^2 + a_2^2 + a_3^2 = |\vec{a}|^2 \text{ — the dot product of a vector with itself gives the **square of its magnitude**.}$$

Properties of the Dot Product

Definition 2: Properties

$$1. \vec{a} \cdot \vec{a} = |\vec{a}|^2$$

$$2. \vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a} \quad (\text{commutative})$$

$$3. \vec{a} \cdot (\vec{b} + \vec{c}) = \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c} \quad (\text{distributive})$$

$$4. (c\vec{a}) \cdot \vec{b} = c(\vec{a} \cdot \vec{b}) = \vec{a} \cdot (c\vec{b})$$

$$5. \vec{0} \cdot \vec{a} = 0$$

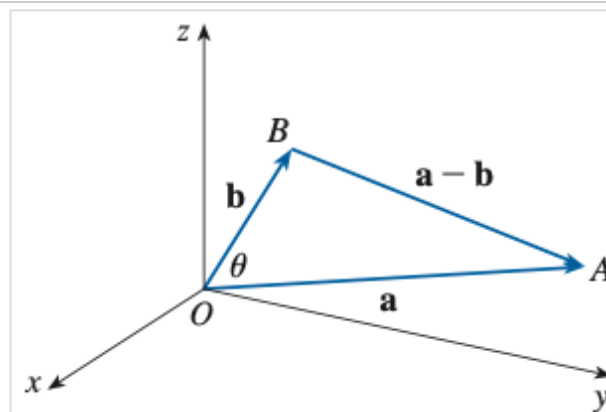
These follow directly from the component definition. Property 1 is especially useful — it connects the dot product to length.

The Geometric Meaning — Angles

Definition 3: Geometric Form of the Dot Product

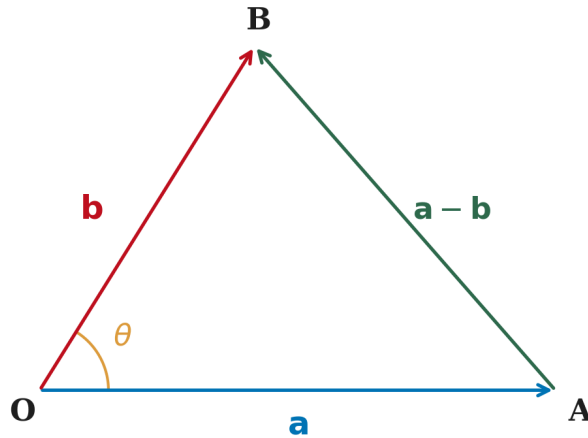
If θ is the angle between $\vec{a}$ and $\vec{b}$ (with $0 \leq \theta \leq \pi$), then

$$\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta$$



This connects **algebra** (the component formula) with **geometry** (the angle between vectors)!

$$\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|}$$



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Example: Finding the Angle Between Vectors

Example 2

Find the angle between $\vec{a} = \langle -1, 3, 4 \rangle$ and $\vec{b} = \langle 5, 2, 1 \rangle$.

Compute a dot b , then $|a|$ and $|b|$, then use the angle formula.

Orthogonality

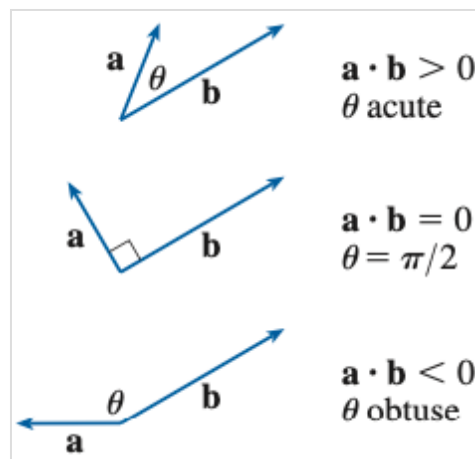
Definition 4: Orthogonal Vectors

Two vectors $\vec{a}$ and $\vec{b}$ are **orthogonal** (perpendicular) if and only if

$$\vec{a} \cdot \vec{b} = 0$$

Why? Because $\cos 90^\circ = 0$, so $\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cdot 0 = 0$.

Sign of $\vec{a} \cdot \vec{b}$	Angle between vectors
$\vec{a} \cdot \vec{b} > 0$	Acute ($\theta < 90^\circ$)
$\vec{a} \cdot \vec{b} = 0$	Right angle ($\theta = 90^\circ$)
$\vec{a} \cdot \vec{b} < 0$	Obtuse ($\theta > 90^\circ$)



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This is the **most important** use of the dot product — a quick test for perpendicularity!

Your Turn — Orthogonality Check

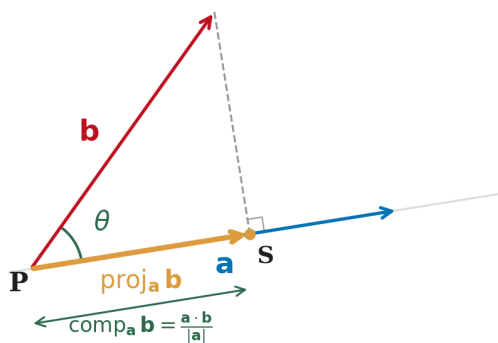
Example 3

Are $\vec{u} = \langle 2, 1, -1 \rangle$ and $\vec{v} = \langle 1, -1, 1 \rangle$ orthogonal?

Compute $u \cdot v$ and check if it equals zero.

Projections

How much of $\vec{b}$ points in the direction of $\vec{a}$?



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Definition 5: Scalar and Vector Projections

Scalar projection (component of $\vec{b}$ onto $\vec{a}$):

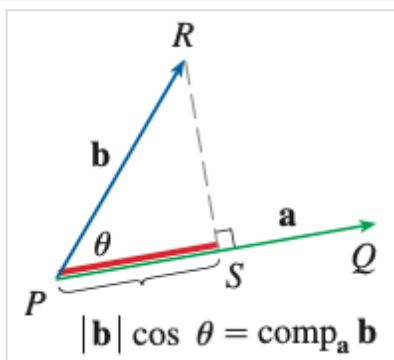
$$\text{comp}_{\vec{a}} \vec{b} = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}|}$$

This is a **number** — the signed length of the shadow.

Vector projection of $\vec{b}$ onto $\vec{a}$:

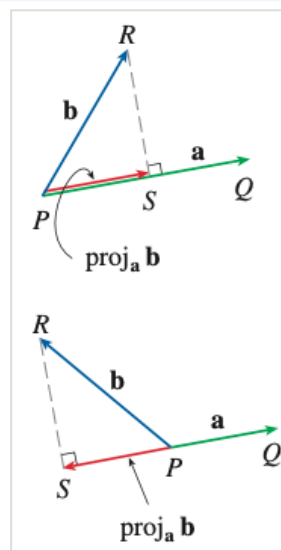
$$\text{proj}_{\vec{a}} \vec{b} = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}|^2} \vec{a}$$

This is a **vector** — the actual shadow vector along $\vec{a}$.



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comp = how far, proj = the actual vector



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Example: Computing Projections

Example 4

Find the scalar and vector projections of $\vec{b} = \langle 12, 1, 2 \rangle$ onto $\vec{a} = \langle -1, 4, 8 \rangle$.

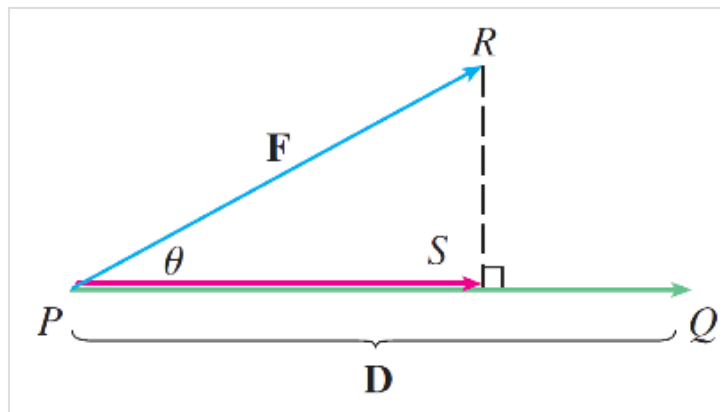
Compute a dot b, then $|\vec{a}|$, then use the projection formulas.

Work — A Physical Application

When force is aligned with motion: $W = |\vec{F}| \cdot |\vec{D}|$

When force is at an angle θ to the motion:

$$W = \text{comp}_{\vec{D}} \vec{F} \cdot |\vec{D}| = \vec{F} \cdot \vec{D}$$

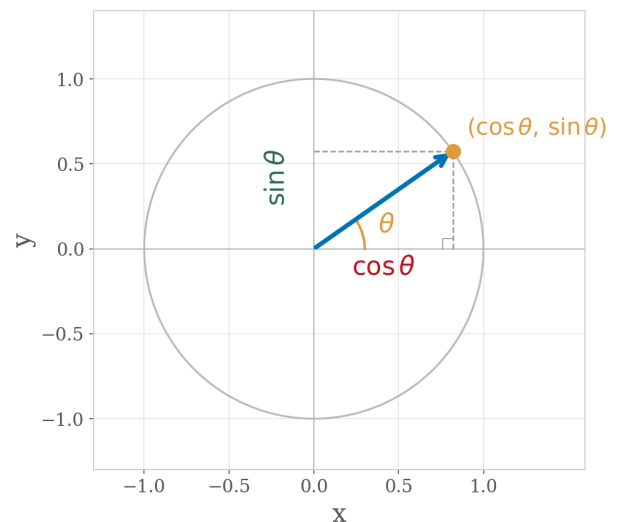


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Example 5

A wagon is pulled 100 m by a force of 70 N applied at 35° above horizontal. Find the work done.

Set up the force and displacement vectors, then compute $F \cdot D$.



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$$\vec{F} = 70\langle \cos 35^\circ, \sin 35^\circ \rangle, \quad \vec{D} = \langle 100, 0 \rangle$$

$$W = \vec{F} \cdot \vec{D} = 70 \cos 35^\circ \cdot 100 + 70 \sin 35^\circ \cdot 0 = 7000 \cos 35^\circ \approx 5734 \text{ N}\cdot\text{m}$$

Your Turn — Work

Example 6

A box is pushed 5 m along the floor by a 40 N force applied at 20° below horizontal. Find the work done.

Set up the force and displacement vectors, then compute $F \cdot D$.

Homework